

Laxima Niure Kandel, Ph.D.

Assistant Professor (Tenure-Track)
Director, Cybersecurity Engineering Laboratory
Director, CyBASE Center
Director, Kandel Research Lab
Electrical Engineering and Computer Science Department
Embry-Riddle Aeronautical University (ERAU)
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<https://sites.google.com/view/laximakandel/home> Kandel Research Group | ✉ niurekal@erau.edu | 🏠 <https://faculty.erau.edu/Laxima>

RESEARCH AREAS

Security & Privacy

- UAV Security
- GPS Spoofing and Jamming
- Differential Privacy & Federated Learning

MIMO RF-Signal Processing

- AI-enabled Cybersecurity
- Wireless Sensing & Imaging
- Indoor Localization

EDUCATION

Ph.D. in Computer Engineering

Stevens Institute of Technology, Hoboken, NJ

Dissertation: Exploiting Commodity WiFi Devices for Continuous Spatial Awareness and Security

Advisor: Prof. Shucheng Yu

Outstanding Dissertation Award

July 2020

MSc. in Telecommunication and Information Systems

University of Essex, Colchester, UK

Advisor: Prof. Kenneth Guild

Nov 2008

BEng. in Electronics and Telecommunication

Priyadarshini College of Engineering and Architecture, Nagpur, India

Advisor: Prof. N.S. Ambatkar

June 2007

EMPLOYMENT HISTORY

- | | |
|--|-------------------------------------|
| • Assistant Professor (Tenure-Track),
EECS Department, ERAU, FL | August 2020 - Present |
| • Director, Cybersecurity and Assured Systems
Engineering Center (CyBASE), ERAU | Fall 2024-Present |
| • Director, Cybersecurity Engineering Laboratory, ERAU | Summer 2023-Present |
| • Graduate Teaching Assistant, Stevens Institute of Technology, NJ | Fall 2017, Spring 2018, Spring 2020 |
| • Graduate Research Assistant, Stevens Institute of Technology, NJ | Summer 2018 - Fall 2019 |
| • Lecturer, University of Arkansas at Little Rock, AR | Spring 2015 - Fall 2017 |

TEACHING EXPERIENCE

1. Instructor of Record, Assistant Professor, ERAU

- SYS 690 Systems Engineering Project (Spring 2025)
- CS 427 System Exploitation and Penetration Testing (Spring 2025)

- CS 599 **Special Topics in Computer Science** (Fall 2024)
 - CS 303 **Cryptography & Network Security** (Fall 2024, Fall 2025)
 - CS 538 **Applied Cryptography** (Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025, Spring 2026)
 - CS 428 **Applied Cryptography** (Spring 2021, Spring 2022, Spring 2023, Spring 2025, Spring 2026)
 - CS 426 **Digital Forensics** (Spring 2022, Spring 2023, Spring 2026)
 - CEC 470 **Computer Organization & Architecture** (Fall 2020, Fall 2021, Fall 2022, Fall 2023)
 - SE 599 **Special Topics Software Engineering** (Fall 2022)
 - CS 222 **Intro to Discrete Structures** (Spring 2021)
 - CS 690 **Cybersecurity Engineering Capstone Project** (Spring 2021)
 - CS 700 **Graduate Thesis** (Fall 2020, Spring 2021, Fall 2021, Fall 2024)
 - SE 700 **Graduate Thesis** (Spring 2024, Fall 2024, Spring 2025)
 - CS 800 **Dissertation** (Spring 2024, Fall 2024, Spring 2025)
2. **Teaching Assistant, Stevens Institute of Technology**
- CPE 487 **Digital System Design** (Fall 2017, Spring 2018)
 - CPE 390 **Microprocessor Systems** (Fall 2018, Spring 2020)
3. **Instructor of Record, Lecturer, UALR Computer Literacy (Fall 2015 - Spring 2017)**

RESEARCH FUNDING

1. FUNDED PROJECTS

1.12 Project Title: *"GPS Spoofing and Jamming Attacks on UAVs with Low-Cost SDRs"*

- Role: PI
- Fund: \$19,564
- Agency: COE-BCAAS Research Stimulus Program
- Status: Awarded
- Duration: Nov 2024 - Jul 2025

1.11 Project Title: *"Experiential Accessible Learning Labs for Promoting Inclusive Software Development"*

- Role: PI
- Fund: \$113,905
- Agency: NSF
- Status: Awarded
- Duration: Jul 2024 - Aug 2029

1.10 Project Title: *"NSF REU Site: Cybersecurity of Unmanned Aerial Vehicles"*

- Role: PI
- Fund: \$400,418.00
- Agency: NSF
- Status: Awarded
- Duration: Mar 2023 - Feb 2026

1.9 Project Title: *"Cyber Eagles: High-skilled Workforce Development for the Aviation and Aerospace Cybersecurity Domain"*

- Role: Co-PI
- Fund: \$ 3,875,300
- Agency: National Science Foundation (NSF)
- Status: Awarded
- Duration: Feb 2022 - Jan 2027

1.8 Project Title: “ *Supporting Aviation Cyber Initiative (ACI)* ”

- Role: Subject Matter Expert (SME)
- Fund: \$ 300,000
- Agency: Federal Aviation Administration (FAA)
- Status: Awarded
- Duration: Jan 2024- Dec 2024

1.7 Project Title: “ *Autonomous Airfield Vehicle System (AAV) Cyber Assessment* ”

- Role: Co-PI
- Fund: \$ 150,000
- Agency: NARTP
- Status: Awarded
- Duration: May 2024- Aug 2025

1.6 Project Title: “ *Cybersecurity Data Science (CSDS) FAA Task Order 26* ”

- Role: Co-PI
- Fund: \$ 1,184,951
- Agency: Federal Aviation Administration (FAA)
- Status: Awarded
- Duration: Jun 2023- Aug 2024

1.5 Project Title: “CoE Research and Innovation Stimulus Funds”

- Role: PI
- Fund: \$21,840
- Agency: CoE, ERAU
- Status: Awarded
- Duration: Mar 2023 - June 2023

1.4 Project Title: “Course Design Grant for Information and Computer Security”

- Role: PI
- Fund: \$1500
- Agency: CTLE, ERAU
- Status: Awarded
- Duration: June 2023 - December 2023

1.3 Project Title: “ *Cybersecurity Training Curriculum Development* ”

- Role: Co-PI
- Fund: \$ 974,886
- Agency: Federal Aviation Administration (FAA)
- Status: Awarded
- Duration: Sep 2022 - Sep 2023

1.2 Project Title: “*Expanding the Nation’s STEM Talent Pool through Scholarships to Accelerate Graduates in Engineering*”

- Role: Senior Personnel
- Fund: \$ 1,499,356
- Agency: NSF
- Status: Awarded
- Duration: Oct 2022 - Sep 2027

1.1 Project Title: “*Identifying Unauthorized UAVs using RF Fingerprinting*”

- Role: PI
- Fund: \$26,850

- Agency: FIRST, Internal Grant ERAU
- Status: Awarded
- Duration: May 2021 - Jun 2022

2. SUBMITTED FUNDING PROJECTS [Pending and Unfunded]

2.14 Project Title: “AI for Resilient Manufacturing ”

- Role: Senior Personnel
- Fund: \$2.434613M
- Agency: NIST
- Status: **Pending**
- Submitted date: Jan 15, 2025

2.13 Project Title: “CAREER: RF Fingerprinting-Based UAV Detection and Authentication ”

- Role: PI
- Fund: \$539,708
- Agency: NSF
- Status: **Pending**
- Submitted date: July 23rd, 2025

2.12 Project Title: “REU Site: Cybersecurity of Unmanned Aerial Vehicles ”

- Role: PI
- Fund: \$449,465
- Agency: NSF
- Status: **Pending**
- Submitted date: August 22, 2025

2.11 Project Title: “SaTC 2.0: RES: Dynamic Analog Radio-Frequency Fingerprints (DARF) for Wireless Communications Security ”

- Role: Co-PI
- Fund: \$1.193123M
- Agency: NSF
- Status: **Pending**
- Submitted date: September 29, 2025

2.10 Project Title: “Collaborative Proposal: VINES: Track 1: AeroSense: Airspace-Aware RAN with Reflectarrays and RF Sensing Apparatus for Low-Altitude Drone Identification and Tracking ”

- Role: PI
- Fund: \$300,000
- Agency: NSF
- Status: **Returned**
- Submitted date: August 29, 2025

2.9 Project Title: “Advanced/Alternate Position, Navigation and Timing (PNT) ”

- Role: PI
- Fund: \$2.389622M
- Agency: Space Development Agency (SDA)
- Status: **Unfunded**
- Submitted date: March 15, 2025

2.8 Project Title: “ Research and Develop AI Literacy Toolkit for Training Faculties ”

- Role: Co-PI
- Fund: \$367,877
- Agency: NSF

- Proposal Number: 2518711
- Status: **Unfunded**
- Submitted date: Jan 15, 2025

2.7 Project Title: “ *Mid-scale RI-1 (M1:IP): National Aerospace Cyber Range* ”

- Role: Co-PI
- Fund: 7,942,533
- Agency: NSF
- Proposal Number: 2512291
- Status: **Not Invited**
- Submitted date: Nov 18, 2024

2.6 Project Title: “ *SaTC: EDU: Empowering the Next Generation: AI and ML for Cybersecurity* ”

- Role: Co-PI
- Fund: \$398,945
- Agency: NSF
- Proposal Number: 2427688
- Status: **Unfunded**
- Submitted date: April 08, 2024

2.5 Project Title: “*Research Infrastructure: MRI: Track I Acquisition of a Cybersecurity Resiliency Artificial Intelligence Instrument for Aerospace Systems* ”

- Role: Co-PI
- Fund: \$334,175
- Agency: NSF
- Proposal Number: 2408190
- Status: **Unfunded**
- Submitted date: November 15, 2023

2.4 Project Title: “*Collaborative Research: CPS: Small: NSF-DST: Development of Near Real-time ML-based System for Aviation Applications to monitor threats from Space weather, GNSS spoofing and RFI* ”

- Role: Co-PI
- Fund: \$385,954
- Agency: NSF
- Proposal Number: 2413799
- Status: **Unfunded**
- Submitted date: December 15, 2023

2.3 Project Title: “*CAREER: Securing the Skies with Physical-Layer Security Based UAV Authentication* ”

- Role: PI
- Fund: \$520,489
- Agency: NSF
- Proposal Number: 2340639
- Status: **Unfunded**
- Submitted date: July 26, 2023

2.2 Project Title: “ *ERI: Secure Drone Authentication Using Physical Layer Information* ”

- Role: PI
- Fund: \$197,498
- Agency: NSF
- Proposal Number: 2138600
- Status: **Unfunded**

- Submitted date: June 18, 2021

2.1 Project Title: “*SERVICE: Speed-Up of Exploration of Reliable and Validated AI-Models for Internet-based Solutions for Cyber-Physical E-Health Applications*”

- Role: Senior Personnel
- Fund: 500K Euros (approx. 580K)
- Agency: European Commission
- Status: **Unfunded**
- Submitted date: July 22, 2022

PUBLICATIONS

Book Chapters

- B1. D. Gupta, L. A. Kavuri, L. Elluri, **L. N. Kandel**, and N. Nader, “NextGen Strategies and Tools: Pioneering Innovation in Agriculture”, CRC Press, 2025 (**Accepted**)

Journal Papers

- J5. J. Webb, and **L. N. Kandel**, “Comparative Performance Analysis of Cryptographic Workloads Across Cloud Providers: A Multi-Language Study on FaaS and IaaS Platform”, IEEE Access, 2025 (**Preparation for Resubmission, First Round of Reviews Received**).
- J4. Angara S, **L. N. Kandel**, Dhakal R. Cybersecurity in Smart Grids: A Domain-Centric Review. Systems. 2025; 13(12):1119. <https://doi.org/10.3390/systems13121119>.
- J3. Dhakal R, **L. N. Kandel**, Shekhar P. Radio Frequency Fingerprinting Authentication for IoT Networks Using Siamese Networks. IoT. 2025; 6(3):47. <https://doi.org/10.3390/iot6030047>.
- J2. R. Dhakal, W. Raza, V. Tummala and **L. N. Kandel**, “Enhancing Intrusion Detection in IoT Networks Through Federated Learning,” in IEEE Access, vol. 12, pp. 167168-167182, 2024, doi: 10.1109/ACCESS.2024.3495702.
- J1. Dhakal, Raju, and **L. N. Kandel**. “Physical-Layer-Based UAV Identification: A Comprehensive Review.” Journal of Aerospace Information Systems (2025): 1-19.

Conference Papers

- C29. M. Stefanov, S. Denchev, K. Stefanov, B. Jekov, and **L. N. Kandel**, “Intelligent Detection and Response System for Attacks on Unmanned Aerial Vehicles Based on Hybrid AI Models,” accepted for oral presentation at the 9th International Conference on Information and Computer Technologies (ICICT 2026), Honolulu, Hawaii, USA, Mar. 11–13, 2026. (**Accepted**)
- C28. G. Venkata Ramana, S. Miriyala, **L. N. Kandel**, D. Venkata Ratnam, and K. Deshpande, “Detection and Analysis of Radio Frequency Interference Effects on GNSS Signals and Positioning Accuracy,” in Proc. 2026 National Radio Science Meeting, Boulder, CO, Jan. 6–9, 2026. (**Accepted, see program information** https://www.nrsboulder.org/view_paper.php?PaperNum=1233)
- C27. K. D. Taylor, J. L. Radjavitch, and **L. N. Kandel**, “WIP: Integrating Accessible Learning Labs: an Exploration of Faculty Experiences”, 2025 IEEE Frontiers in Education Conference (FIE), Nashville, Tennessee. (**Accepted**)
- C26. V. Li, J. Villa, J. Clifford, D. Diessner, and **L. N. Kandel**, “C/N₀ analysis-based GPS spoofing detection with variable antenna orientation,” in Proc. IEEE 5th Cyber Awareness and Research Symposium (CARS 2025), Univ. of North Dakota, Grand Forks, ND, USA, 2025. (**Accepted**)(**Received 3rd Best Paper**)
- C25. Dhakal, R., Sekher, P., **Kandel, L. N.** (2025). “Adversarial-resilient RF fingerprinting: A CNN-GAN framework for rogue transmitter detection”. In 24th International Conference on Machine Learning and Applications (ICMLA), Boca Raton, Florida, USA. (**Accepted**)
- C24. Okatan, A. S., Akbaş, M. İ., **Kandel, L. N.**, Pekoz, B. “Keys in the Weights: Transformer Authentication Using Model-Bound Latent Representations”. In IEEE 5th Cyber Awareness and Research Symposium 2025 (CARS), University of North Dakota, Grand Forks, ND, USA. (**Accepted**)
- C23. Okatan, A. S., Akbaş, M. İ., **Kandel, L. N.**, Pekoz, B. “Seed-Induced Uniqueness in Transformer Models: Subspace Alignment Governs Subliminal Transfer”. In IEEE 5th Cyber Awareness and Research Symposium 2025 (CARS), University of North Dakota, Grand Forks, ND, USA. (**Accepted**)

- C22. Gharami K., Moni S. Showkat, **Kandel L.N.**, “Siamese mViT: Lightweight Biometric Facial Recognition for Edge Devices” In IEEE 5th Cyber Awareness and Research Symposium 2025 (CARS), University of North Dakota, Grand Forks, ND, USA. **(Accepted)**
- C21. M. Durieux, K. Taylor, **L. N. Kandel** and D. Gupta, “GPS Spoofing Attacks and Pilot Responses Using A Flight Simulator Environment”, The 7th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications, Pittsburgh, PA, USA, 2025. **(Accepted)**
- C20. B. P. Devkota and **L. N. Kandel**, “Applying Deep Learning Approach for GPS Spoofing Detection of UAV with Explainable AI,” 2025 AIAA DATC/IEEE 44th Digital Avionics Systems Conference (DASC), Montreal, QC, Canada, 2025, pp. 1-8, doi: 10.1109/DASC66011.2025.11257153.
- C19. T. Delbos, K. D. Taylor, **L. N. Kandel** and R. F. Babiceanu, ”Evaluating Pilot Response to ADS-B Attacks Using a Flight Simulator and Low-Cost SDR,” 2025 AIAA DATC/IEEE 44th Digital Avionics Systems Conference (DASC), Montreal, QC, Canada, 2025, pp. 1-9, doi: 10.1109/DASC66011.2025.11257182.
- C18. V. Kulothungan, D. Gupta and **L. N. Kandel**, “Democratizing Cybercrime: Risks and Countermeasures of AI-Enabled Attacks,” 2025 IEEE 11th World Forum on Internet of Things (WF-IoT), Chengdu, China, 2025, pp. 1-6, doi: 10.1109/WF-IoT64238.2025.11270640.
- C17. Stefanov, M., Stefanov, K., **Kandel, L. N.**, Crouse, S., Jekov, B. (2025). “Autonomous Agentic AI architectures for optimizing Security Operations Centers (SOC) KPIS: Methodology, impact on detection, response, and recovery”. Land Forces Academy Review, 30(3), 479–493. <https://doi.org/10.2478/raft-2025-0046>.
- C16. R. Razdan, J. Vargas, I. Akabas, S. O. Alswelss, A. Cheatham, K. Ford, E. Horvath, **L. N. Kandel** and M. Musa, “Product Assurance in the Age of Artificial Intelligence”, No. EPR2025011. SAE Technical Paper, 2025.
- C15. P. Shekhar, B. Devkota, D. Samaraweera, **L. N. Kandel** and M. Babu, “Do Adversarial Patches Generalize? Attack Transferability Study Across Real-Time Segmentation Models in Autonomous Vehicles,” 2025 IEEE Security and Privacy Workshops (SPW), San Francisco, CA, USA, 2025, pp. 322-328, doi: 10.1109/SPW67851.2025.00045.
- C14. L. Luna, M. P. Berkowitz, **L. N. Kandel** and S. Jansen-Sánchez, “Dueling Deep Q-Learning for Intrusion Detection,” SoutheastCon 2025, Concord, NC, USA, 2025, pp. 1192-1197, doi: 10.1109/SoutheastCon56624.2025.10971436. **(Received 1st Best Paper)**
- C13. B. P. Devkota, B. Devkota and **L. N. Kandel**, “Leveraging Tree-Based Machine Learning and Explainable AI for UAV GPS Spoofing Detection,” SoutheastCon 2025, Concord, NC, USA, 2025, pp. 340-345, doi: 10.1109/SoutheastCon56624.2025.10971719.
- C12. Shekhar, P., Devkota, B., Samaraweera, D., **L. N. Kandel**, Babu, M. (2025). “Cross-Model Transferability of Adversarial Patches in Real-time Segmentation for Autonomous Driving”. arXiv preprint arXiv:2502.16012.”
- C11. R. Dhakal, B. P. Devkota and **L. Niure Kandel**, “Radio Frequency Fingerprinting With Siamese Network,” 2025 International Conference on Computing, Networking and Communications (ICNC), Honolulu, HI, USA, 2025, pp. 212-216, doi: 10.1109/ICNC64010.2025.10993537.
- C10. B. P. Devkota, L. Saunders, R. Dhakal and **L. N. Kandel**, “GPS Spoofing Detection with a Random Forest Multiclass Classifier,” MILCOM 2024 - 2024 IEEE Military Communications Conference (MILCOM), Washington, DC, USA, 2024, pp. 202-208, doi: 10.1109/MILCOM61039.2024.10773678.
- C9. P. Choudhary, R. Dhakal and **L. N. Kandel**, “Performance Evaluation of Routing Protocols for UAV- Assisted Post-Disaster Communication Networks,” SoutheastCon 2024, Atlanta, GA, USA, 2024, pp. 1080-1087, doi: 10.1109/SoutheastCon52093.2024.10500104.
- C8. W. Carpenter, P. Omodara, R. Babiceanu and **L. N. Kandel**, “An Extended Study of the Applications of Using Gesture Control to Pilot UAVs,” 2023 International Conference on Machine Learning and Applications (ICMLA), Jacksonville, FL, USA, 2023, pp. 1746-1752, doi: 10.1109/ICMLA58977.2023.00265.
- C7. R. Dhakal, C. Bosma, P. Chaudhary and **L. N. Kandel**, “UAV Fault and Anomaly Detection Using Autoencoders,” 2023 IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), Barcelona, Spain, 2023, pp. 1-8, doi: 10.1109/DASC58513.2023.10311126.
- C6. R. Dhakal and **L. N. Kandel**, “A Survey of Physical Layer-Aided UAV Security,” 2023 Integrated Communication, Navigation and Surveillance Conference (ICNS), Herndon, VA, USA, 2023, pp. 1-8, doi: 10.1109/ICNS58246.2023.10124288.
- C5. X. Xue, M. K. Hasan, S. Yu, **L. N. Kandel** and M. Song, “Over-the-Air Federated Learning with Enhanced Privacy,” ICC 2023 - IEEE International Conference on Communications, Rome, Italy, 2023, pp. 4546-4551, doi: 10.1109/ICC45041.2023.10278765.

- C4. **L. N. Kandel**, Z. Zhang and S. Yu, “Exploiting CSI-MIMO for Accurate and Efficient Device Identification,” 2019 IEEE Global Communications Conference (GLOBECOM), Waikoloa, HI, USA, 2019, pp. 1-6, doi: 10.1109/GLOBECOM38437.2019.9014191.
- C3. **L. N. Kandel**, Z. Zhang and S. Yu, “Poster: Using Commodity WiFi Devices For Object Sensing And Imaging,” 2019 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN), Newark, NJ, USA, 2019, pp. 1-2, doi: 10.1109/DySPAN.2019.8935637.
- C2. **L. N. Kandel** and S. Yu, “VWAN: Virtual WiFi ANtennas for Increased Indoor Localization Accuracy,” 2019 IEEE International Conference on Industrial Internet (ICII), Orlando, FL, USA, 2019, pp. 258-267, doi: 10.1109/ICII.2019.00052.
- C1. **L. N. Kandel** and S. Yu, “Indoor Localization Using Commodity Wi-Fi APs: Techniques and Challenges,” 2019 International Conference on Computing, Networking and Communications (ICNC), Honolulu, HI, USA, 2019, pp. 526-530, doi: 10.1109/ICCNC.2019.8685501.

AWARDS & HONORS

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- | | |
|---|-----------|
| • 2025 College of Engineering Outstanding Teaching Award | 2025 |
| • Nominated for ERAU Undergraduate Research Mentorship Award | 2025 |
| • Served as the panelist for the ICAO-Industry Consultative Forum (ICF), fourth meeting (5th and 6th February 2024) | |
| • Selected for NASA JPL-ERAU Academic Exchange Program | Mar. 2023 |
| • Faculty scholarship to attend the Women in CyberSecurity (WiCyS) Conference, Denver, CO. | Mar. 2023 |
| • Outstanding Ph.D. Dissertation Award in Computer Engineering
Stevens Institute of Technology. | Jul. 2020 |
| • NSF Student Travel Grant, IEEE Globecom | Dec. 2019 |
| • Stevens Ph.D. Conference Funding, Stevens Institute of Technology | Sep. 2019 |
| • Outstanding Teaching Assistant Award , Stevens Institute of Technology | Jan. 2018 |
| • Selected to Participate in the Student Interview Panel for the Program Administration,
UALR | Mar. 2017 |
| • Completed Master’s Degree with Merit from Essex University Colchester, UK | Nov. 2008 |
| • Top 5 from 200 Students in a Campus Interview for AZTECSOFT (Software Company), India | Mar. 2006 |
| • Letter of Appreciation for Securing 4th Position in Nagpur University, India | Jan. 2002 |

SERVICE, OUTREACH & SYNERGISTIC ACTIVITIES

2020-present

1. Ph.D. Dissertation Committee Chair, EECS Department

- Dissertation Title: “*UAV Security Leveraging the Physical Layer Approach*” (Student: Raju Dhakal)
Expected Graduation Fall 2026
- Dissertation Title: “*GPS Spoofing Detection and Mitigation Using Machine Learning Approaches*”(Student: Bhawana Poudel)
Expected Graduation Spring 2028
- Dissertation Title: “*ML/AI based Design and Monitoring of Algae Blooms in Florida*”(Student: Kayla Taylor)
Expected Graduation Spring Spring 2029

2. Ph.D. Dissertation Committee Member, EECS Department

- Dissertation Title: “*Additively Manufactured Antennas for CubeSat Applications*” (Student: Blake Roberts)
Expected Graduation Spring 2026
- Dissertation Title: “*Classification of Electrical Faults and Power Quality Disturbances in Aerospace Power Systems Using Deep Learning*” (Student: Ian C. Guzman)
Expected Graduation Spring 2026
- Dissertation Title: “*Generating and Validating Digital Signatures through Machine Learning*” (Student: Juan Ortiz Couder)
Expected Graduation Fall 2025

- Dissertation Title: “*A Complete Scenario-Based Testing Framework for the Exploration and Exploitation of Autonomous Vehicle Validation Scenarios*” (Student: Quentin Goss)
Expected Graduation Fall 2025
- Dissertation Title: “*Enhancing Security of Proof-of-Learning Against Spoofing Attacks Using Model Watermarking Techniques*” (Student: Ural Ozgur)
Expected Graduated Summer 2025
- Dissertation Title: “*Machine Learning for Intrusion Detection in the 6G UAV Flying Base Stations*”(Student: Faisal Alrefaei)
Graduated Spring 2024
- Dissertation Title: “*Deep CNN-Based Automated Optical Inspection for Aerospace Components*” (Student: Shashi Bhushan Jha)
Graduated Fall 2023
- Dissertation Title: “*Machine Learning and Artificial Intelligence Methods for Anomaly Detection in the Aviation Cybersecurity Ecosystem*” (Student: Anna Baron Garcia)
Graduated Fall 2022
- Dissertation Title: “*Physical-Layer Security Enhancement of Wireless Communications Systems Using Antenna-based RF Fingerprinting Techniques*” (Student: Noemi Miguelez Gomez)
Graduated Fall 2022
- Dissertation Title: “*Supporting the Discovery, Reuse, and Validation of Cybersecurity Requirements at the Early Stages of the Software Development Lifecycle*” (Student: Jessica Steinmann)
Graduated Fall 2022

3. M.S. Thesis Chair, EECS Department, ERAU

- Thesis Title: “*Comparative Performance Analysis of Cryptographic Workloads Across Cloud Providers: A Multi-Language Study on FaaS and IaaS Platforms*”. (Student name: Jeremiah Webb, graduated Spring 2025)
- Thesis Title: “*Performance Evaluation of Routing Protocols and implementation of ML based energy efficient routing protocols for UAV-Assisted Post-Disaster Communication Networks*”(Student: Prachi Choudhary, graduated Fall 2024)
- Thesis Title: “*Towards a Federated Identity and Access Management Across Universities*” (Student: Jameel Alsulami, graduated Fall 2021)
- Thesis Title: “*RF Fingerprinting Unmanned Aerial Vehicles*”(Student: Norah Ondus, graduated Fall 2021)
- Thesis Title: “*A Framework to Detect the Susceptibility of Employees to Social Engineering Attacks*” (Student: Hashim H. Alneami, graduated Fall 2021)

4. M.S. Thesis Member, EECS Department, ERAU

- Thesis Title: “*Study of Output and Behavior of LLMs using Confidence Framing in Prompt Engineering*” (Student name: Micah Parrilla, expected graduation Spring 2026)
- Thesis Title: “*Cybersecurity Risks and Defense Strategies for ML-Powered Future Autonomous Weapon Systems*” (Student name: Thomas J Neubert, Expected Graduation Spring 2026)
- Thesis Title: “*In the Shadow of Prompts: Adversarial Attacks and Model Cloning in Large Language Models*” (Student name: Kanchon Gharami, graduated Summer 2025)
- Thesis Title: “*Applying Natural Language Processing to Software Verification*” (Student name: Luke Newcomb, graduated Spring 2025)
- Thesis Title: “*Machine-Learning Based Strategies to GPS Spoofing Detection*”(Student: Alberto Squatrito, graduated Spring 2024)
- Thesis Title: “*A System for the Detection of Adversarial Attacks in Computer Vision via Performance Metrics*”(Student: Sarah Reynolds, graduated Summer 2023)

4. Service, ERAU

- Panalist, Pushing Boundaries with Artificial Intelligence, ERAU Admissions Panel Spring 2026
- EECS PhD Operating Committee Spring 2026-Present

- Point of Contact (POC) for NSA CAE-CD for ERAU, Daytona Beach Campus Spring 2024-Present
- Discovery Day Judge Spring 2025
- Faculty Mentor for Junior Faculty Shafika Moni, ERAU Fall 2024-Present
- Panalist, Aviation Cyber Expert Panel, ERAU Admissions Panel Spring 2024
- EECS Department Chair Search Adhoc Committee, ERAU Summer 2023,2025
- Faculty Search Committee (External Member), College of Arts and Sciences, ERAU Fall 2022-Present
- Faculty Guide to “Peer Network” program, ERAU Fall 2022-Spring 2023
- Planning Committee, ERAU Spring 2022-Present
- Diversity, Equity, Belonging, and Inclusion (DEBI) Campus-Wide Committee, ERAU Fall 2023-Present
- COE-DEI (Diversity, Equity and Inclusive Committee), College of Engineering, ERAU Fall 2021-Present
- Research award committee, College of Engineering, ERAU Spring 2022
- Faculty Search Committee Member, EECS Department, ERAU Spring 2021, Fall 2021-Spring 2022, Fall 2023, Spring 2023
- Judge for Undergraduate Research, ERAU Spring 2021, Fall 2021
- Oral Ph.D. Qualifying Exam Committee, EECS Department, ERAU Spring 2021, 2022, 2023, 2024

5. Service, External

- Organized Cyber Safety for Seniors Workshop at Port Orange Regional Library April 2025
- Mentor, WiCyS Mentoring Program 2024-2025
- Vice-chair, IEEE Daytona WIE Affinity Group Fall 2022-Present

6. Outreach (K-12 Summer Camp), ERAU

Fundamentals of ML/AI: Project Based Introduction Summer 2026

- Week-long summer program teaching fundamentals of machine learning (ML) and artificial intelligence (AI) to high school students ages 14-18
- Hands-on learning through engaging, project-based activities

Basic cybersecurity summer camp instructor Summer 2022, 2023, 2024, 2025, 2026

- Week-long summer program teaching cybersecurity concepts to high school students ages 14-18
- Hands-on learning of commonly exploited vulnerabilities

7. Coach for Collegiate-focused Cybersecurity Competition, ERAU

- National Cyber League Fall 2025
- NCAE Cyber Games (**Southeast 1 Regional 2025: Team 1 in 1st Position, Team 2 in 6th Position**)
- National Transportation Cybersecurity Competition (NTCC), Fall 2024 (**Dev Patel in 3rd Position**)
- Hivestorm 2022
- International Cyber League Collegiate Cup 2022

8. Student Clubs

- Faculty Advisor for Women in Cybersecurity (WiCyS) club, ERAU Fall 2024-Present
- Faculty Advisor for White Hat Eagles Club (WHEC), ERAU Fall 2024-Present
- Faculty Advisor for Cryptocurrency Club, EECS Department, ERAU Spring 2021-Present
- Faculty Advisor for Upsilon Pi Epsilon Club , ERAU Fall 2022-Present

8. Conference Planning Committee, Stevens

- 15GRC:15th Annual Graduate Research Conference April 30-May 1,2020
Stevens Institute of Technology,Hoboken,NJ

STUDENTS SUPERVISED

PhD Dissertation Chair, ERAU

- Student: *Raju Dhakal* Fall 2022 - Present
Thesis Title: "UAV Security Leveraging the Physical Layer Approach"
- Student: *Kayla Taylor* Spring 2025 - Present
Thesis Title: "ML/AI based Design and Monitoring of Algae Blooms in Florida"
- Student: *Bhawana Paudel* Spring 2024 - Present
Thesis Title: "GPS Spoofing Detection and Mitigation Using Machine Learning Approaches"

M.S. Thesis Chair, ERAU

- Student: *Jeremiah Webb* 2024-2025
Thesis Title: "Comparative Performance Analysis of Cryptographic Workloads Across Cloud Providers: A Multi-Language Study on FaaS and IaaS Platform"
- Student: *Prachi Chaudhary* 2023 - 2024
Thesis Title: "Performance Evaluation of Routing Protocols and implementation of ML based energy efficient routing protocols for UAV-Assisted Post-Disaster Communication Networks"
- Student: *Hashim Alneami* 2020 - 2021
Thesis Title: "A Framework to Detect the Susceptibility of Employees to Social Engineering Attacks"
- Student: *Jameel Alsulami* 2020 - 2021
Thesis Title: "Towards a Federated Identity and Access Management Across Universities"
- Student: *Norah Ondus* 2020 - 2021
Thesis Title: "RF Fingerprinting Unmanned Aerial Vehicles"

M.S. Graduate Research Project (GRP), ERAU

- Student: *Sivasankar Sibi* Spring 2026
GRP title: "Design and Validation of a GNSS Resilience Testbed for UAV Navigation Security"
- Student: *Sahithi Angara* Spring 2025
GRP title: "Systems Engineering Principles for the Modern Smart Grid: DER Integration and Cybersecurity Perspective"
- Student: *Maisa Alghamdi* Spring 2021
GRP title: "Secure Angle: Improving Wireless Security Using Angle-of-Arrival Information"
- Student: *Mohammed Al-Buhayri* Spring 2021
GRP title: "Numerating easy implementation steps for ISO 27001"

Senior Design Product Owner, ERAU

- Students: *Anthony Wirsing, Tyler Covell, Tiffany Flaherty, Dev Patel, Hayley Oo* Fall 2025-Present
Project title: "Development and Security Evaluation of a Portable Aircraft OEM Cell for ICS Cybersecurity Training"

Undergraduate Students, ERAU

- *Asmitha Muthukumar and Katelyn Merritt* Fall 2025-Present
Research: Simulate social engineering attacks and evaluate the most effective attack methods and susceptible victims. Evaluate the effectiveness of AI-generated vs human-written phishing emails.
- *Justin Villa* Fall 2024-Present
Research: GPS Spoofing Detection via Alltitude-Aware SNR Analysis
- *Lucas Saunders, Cross Francis* Fall 2023-Present
Research: GPS Spoofing and Jamming Attacks on UAVs with Low-Cost SDRs
- *Santos Jeorge, Mark Frees & Alessandra Oo* Spring 2022-Spring 2023
Research: Identifying Unauthorized UAVs using RF Fingerprinting
- *Carly Bosma, Olivia Meholic, Call Robinson, Gianna M. Scarangelli* Fall 2022 -Spring 2024

PEDAGOGICAL & CAREER TRAINING

- Joined Asynchronous Short Course for Faculty on Artificial Intelligence (AI), Center for Teaching and Learning Excellence (CTLE), ERAU-DB Fall 2025
- Participant, Open Platform Infrastructure-Industrial Control Systems Security (OPI-ICS) Faculty Development Workshop Summer 2025
- Participant, AI-Assisted Cybersecurity Faculty Workshop Summer 2025
- Panelist for the ICAO-Industry Consultative Forum (ICF), Fourth Meeting Spring 2024
- Participated in Maymester: Inclusive Teaching Materials (2-day CTLE workshop @ERAU) Spring 2023
- Participant in Community of Practice-Inclusive Teaching at ERAU-DB Fall 2022-Spring 2023
- Summer Grant Writing Workshop Program Organised by University of South Florida, Sarasota July 2021
- Pivotal Pedagogy Seminars, CTLE, ERAU July 2020, Feb 2022
- Teaching at the College Level, Stevens Institute of Technology Fall 2019
 - Participated in semester long workshop introducing how learning works and best practices for effective teaching
- Writers Group, Stevens Institute of Technology Fall 2017
 - Completed an 8-week workshop for improving technical and professional writing skills

LICENCES & CERTIFICATION

- Scrum for Managers Oct 2020(Issued)-No Exp.
- Scrum for Product Owners Oct 2020(Issued)-No Exp.
- Scrum for Team Members Oct 2020(Issued)-No Exp.

SOFTWARE SKILLS

Programming Languages: C, C++, JAVA, Python, MATLAB, LATEX
 Hardware Description Language: VHDL
 Database Systems: MS SQL Server, MS Access
 Operating Systems: Windows, Linux
 Applications: Visual C++, Wireshark, Microsoft Office, GNU Radio, Xilinx Tanner
 Learning System: Canvas, Blackboard, Perusal, Nearpod

PROFESSIONAL ACTIVITIES

- Publicity Chair, IEEE International Conference on E-health Networking, Application and Services, New York, USA <https://healthcom2026.ieee-healthcom.org/committees>
- NSF Panelist 2023, 2026
- SoutheastCon Session Chair; TPC Member 2024
- DASC 2025 Session Chair (CS1 - Security Analysis and Modelling; A15: Certification Implementation) & TPC Member 2025
- ICC TPC Member 2023, 2024
- IEEE LNC TPC Member, 2023, 2024
- **Technical reviewer:** IEEE Transactions on Aerospace and Electrical Systems, 2025; IEEE Transaction on Vehicular Technology (TVT)-2022, 2023, 2025; IEEE Transaction on Mobile Computing-2020, 2021, 2023; IEEE Systems Journal 2019; SecureComm 2019; IEEE COINS 2022; IEEE CNS 2017,

PROFESSIONAL MEMBERSHIP

IEEE Member
IEEE Aerospace & Electronic Systems Society Membership
IEEE Comsoc Member
IEEE Young Professionals
IEEE Women in Engineering
N2 Women Member
WiCyS Member